Investigating Pipeline Hazards

In my last article, I explained the concept of pipelining, and the various stages involved. We saw the difference between a pipelined and a non-pipelined processor, and how the instructions are executed in a pipeline to increase the throughput. In this article, we will explore the concept of pipeline hazards.

We are often faced with a situation that prevents the next instruction in the instruction stream from being executed during its designated clock cycle. This is known as a pipeline hazard. As shown in Figure 1, we have two instructions in the pipeline. There happens to be a pipeline hazard, which prevents the execution of Instruction 'i1' in the desired clock cycle.

Hazards reduce the performance from the ideal speed-up gained by pipelining. How this happens is explained in the following sections.

Types of pipeline hazards

Pipeline hazards are classified into three types:

- Data hazard
- Structure hazard
- Control hazard

Data hazard

A data hazard occurs when an instruction depends on the result of the preceding instruction in a pipeline. As an example, consider the following instructions:

```
ADD R1, R2, R3 //Instruction i0
MUL R4, R1, R2 //Instruction i1
```

Note: In the above example, let us assume that the first operand is the destination operand, and the remaining are the source operands.

Now, as we can see in the above example, the *'ADD'* instruction updates the value of register *R1*, and the *'MUL'* instruction uses the value of *R1* as updated by the *ADD* instruction. As I discussed in my last article (see the first item in *References*), the updating of the register happens in the write-back (WB) stage. Thus, the *ADD* instruction will update the *R1* register in the WB stage. But the *MUL* instruction will read its source operand in the ID stage itself. Thus, the *MUL* instruction may pick up the wrong value from *R1*, since the WB of the previous instruction has not happened by the time the ID of the current instruction happens. This situation is known as a data hazard. It is often dealt with by either stalling or data forwarding, or a combination of both. I will explain these terms later in this article.

Structure hazard

A structure hazard occurs when a hardware resource conflict occurs between the preceding instruction in a pipeline, and the current instruction. This type of hazard occurs when